Natural and Processed Materials

Answer all of the questions in the spaces below.

1. Positive ions are created when an atom _____ electrons.

2. How many valence electrons do the following elements have?

Na	_____	Cl	_____
Mg	_____	Al	_____
S	_____	Ar	_____
O	_____	P	_____

3. For the elements above in Q 2 state whether the atoms would prefer to **gain or lose electorns** and **how many**. What would be the **formula** of the ion that is created by losing or gaining electrons. Complete the table below.

	Gain/Lose	Number	Formula
Na			
Mg			
S			
O			
Cl			
Al			
Ar			
P			

4. To make stable ions or compounds, elements try to achieve _____ electrons in their outer shells.
Why does this happen?